NSERC CREATE Grant Application

Program: Collaborative Research and Training Experience (CREATE)

Project Title: Advanced Training in Next-Generation MRI: Commercialization of 40 Novel Pulse Sequences for Healthcare and Geospatial Sensing

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1. Executive Summary

This NSERC CREATE program aims to bridge the gap between theoretical quantum magnetic resonance research and clinical/geospatial application. By leveraging a comprehensive library of 40 proprietary, mathematically optimized MRI and NMR pulse sequences—ranging from statistical manifold distributions for dementia tracking to hyperpolarized reconstructions and quantum measure theory for geological oil/ore reservoirs—this program will train the next generation of Highly Qualified Personnel (HQP). Trainees will gain interdisciplinary expertise in quantum physics, software engineering, RF hardware, and commercial translation.

2. Excellence of the Team of Researchers

Led by Cartik Sharma and the NeuroMorph ecosystem, the applicant pool represents a unique convergence of quantum computing, medical biophysics, and mathematical modeling. The team has demonstrated exceptional ability in generating high-impact algorithmic sequences for complex neuromodulation (e.g., Deep Brain Stimulation stage gating via Radon-Nikodym mapping) and geospatial Continued Fraction mapping.

3. Merit of the Proposed Training Program

The core of the program will revolve around the testing, validation, and eventual health/industrial qualification of 40 distinct pulse sequences. Trainees will engage in advanced laboratory and computational modules:

- **Pulse Sequence Programming:** Hands-on development using custom simulators, optimizing RF sweeps and gradients.
- **Quantum Machine Learning (QML):** Translating QML Ansatz models and continued fraction mathematics into actionable imaging and modulation protocols.
- **Clinical and Geospatial Translation:** Porting sequences to identify mineral/hydrocarbon reservoirs or neural degenerative anomalies like Tau and Amyloid-beta oligomers.

4. Need for Highly Qualified Personnel (HQP)

Canada's advanced sensing and medical imaging sectors face a critical shortage of engineers and physicists capable of bridging abstract mathematical concepts with radiofrequency (RF) hardware implementation. This CREATE program addresses the national deficit by generating industry-ready innovators fully proficient in creating deployable IP around next-generation sensors.

5. Program Management and Sustainability

The program will enforce a strict, multi-stage gating process for sequence validation. Intellectual property generated surrounding the 40 pulse sequences will be co-licensed to partner medical facilities and natural resource coalitions, ensuring long-term sustainability of the CREATE training ecosystem and continuing Canadian leadership in quantum sensing.